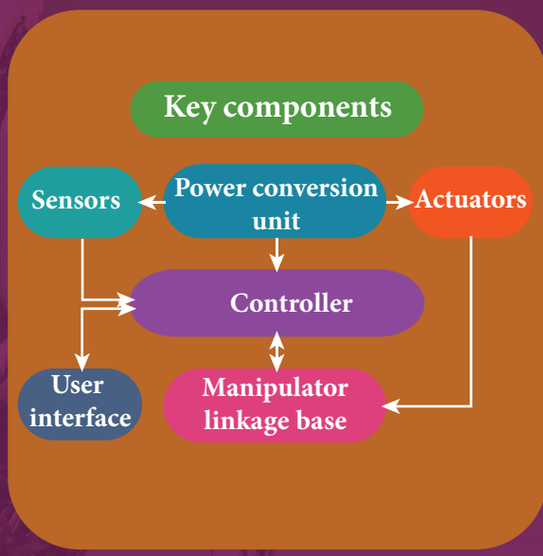
11.3.2 COMPONENTS OF ROBOTICS

The robotic system mainly consists of sensors, power supplies, control systems, manipulators and necessary software.

Most robots are composed of 3 main parts:

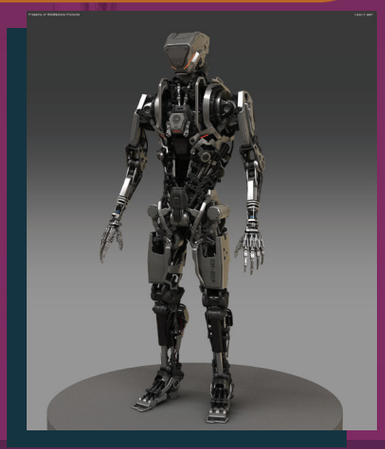
1. **The Controller** - also known as the "brain" which is run by a computer program. It gives commands for the moving parts to perform the job.
2. **Mechanical parts** - motors, pistons, grippers, wheels, and gears that make the robot move, grab, turn, and lift.
3. **Sensors** - to tell the robot about its surroundings. It helps to determine the sizes and shapes of the objects around, distance between the objects, and directions as well.



11.3.3 TYPES OF ROBOTS

HUMAN ROBOT

Certain robots are made to resemble humans in appearance and replicate the human activities like walking, lifting, and sensing, etc.



1. Power conversion unit: Robots are powered by batteries, solar power, and hydraulics.
2. Actuators: Converts energy into movement. The majority of the actuators produce rotational or linear motion.
3. Electric motors: They are used to actuate the parts of the robots like wheels, arms, fingers, legs, sensors, camera, weapon systems etc. Different types of electric motors are used. The most often used ones are AC motor, Brushed DC motor, Brushless DC motor, Geared DC motor, etc.
4. Pneumatic Air Muscles: They are devices that can contract and expand when air is pumped inside. It can replicate the function of a human muscle. They contract almost 40% when the air is sucked inside them.
5. Muscle wires: They are thin strands of wire made of shape memory alloys. They can contract by 5% when electric current is passed through them.
6. Piezo Motors and Ultrasonic Motors: Basically, we use it for industrial robots.
7. Sensors: Generally used in task environments as it provides information of real-time knowledge.
8. Robot locomotion: Provides the types of movements to a robot. The different types are
   (a) Legged (b) Wheeled (c) Combination of Legged and Wheeled Locomotion
   (d) Tracked slip/skid

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